Transistors & relays

National 5 · Electronics & control · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe how a transistor works as a switch (about 0.7 V base–emitter)
- ☐ explain the purpose of a relay and a protection diode
- ☐ choose a transistor to switch a high-current output such as a lamp or motor

I DO — Worked example

Read how a full-mark answer is built

A microcontroller output pin cannot supply enough current to light a 12 V lamp.

1 Name the component that should be added to switch on the lamp. (1 mark)

Points that earn marks from the marking instructions: one ✓ per point

✓ Transistor

Why marks are not awarded

✗ Resistor, relay on its own

WE DO — Guided practice

Use the starters and word bank to help you

A transistor drives a relay coil. A diode is connected across the coil.

2 Explain the purpose of the diode. (2 marks)

Sentence starters

One reason is … because …

Another reason is … so …

Word bank

protecting · transistor · damage · produces · conducts · voltage · switches · safely

Tip: 2 marks = 2 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which component lets a 5 V circuit switch a 230 V heater?

☐ LDR ☐ Buzzer ☐ Diode ☐ Relay

b) Why is a diode placed across a relay coil?

☐ To light up when the relay is on ☐ To increase the current through the coil ☐ To protect the transistor from reverse voltage ☐ To switch the relay on and off faster

c) An NPN transistor switches on at a base voltage of about…

☐ 0.07 V ☐ 0.7 V ☐ 7 V ☐ 70 V

4 Key terms: write the letter of the matching description next to each term.

1. **Transistor switch-on voltage**	____	**A** Lets a low-current circuit switch a separate high-current circuit.
2. **Base resistor**	____	**B** Limits base current to protect the transistor.
3. **Relay**	____	**C** About 0.7 V at the base.

A relay is used between a transistor and a mains lamp.

5 Explain why a relay is used. (2 marks)

6 Explain what is meant by ‘Protection diode’. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?